Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: *DRBENCHMARK* | ***Context:*** *EU Pharmaceutical Regulatory NLP — French*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input form is largely well-aligned. Both deployment and benchmark operate on French text in Latin script with standard diacritics, and the benchmark filters multilingual datasets to French-only [Q31]. No script, modality, or infrastructure mismatch exists for the core deployment. Two concerns reduce the score from 5: (1) sentence-splitting of long EMEA documents [Q37] introduces structural artifacts (QUAERO-D19, QUAERO-D20) and removes document-level context that SmPC/CTD reasoning may require; and (2) PxCorpus introduces transcribed speech with ASR artifacts (PXCORPUS-D10, PXCORPUS-D12, PXCORPUS-D13) that are absent from the deployment's text-only document-management input modality. The benchmark also exposes non-French configs in E3C and MANTRAGSC HuggingFace repositories (E3C-D8, E3C-D9) that could contaminate evaluation if pipelines do not filter correctly. Tokenization variation is studied [Q78–Q80] and shown to be a minor effect.

ID	Evidence
Q31	“While the dataset covers 5 languages, only the French portion is retained for the benchmark.” (p.3)
Q37	“Due to the presence of nested entities in annotations, we simplified the evaluation process by retaining only annotations at the higher granularity level from the BigBio (Fries et al., 2022) implementation, following the approach described in Touchent et al. (2023), which translates into an average loss of 6.06% of the annotations on EMEA and 8.90% on MEDLINE. Additionally, considering that some documents from EMEA exceed the maximum input sequence length that most current language models can handle, we decided to split these documents into sentences.” (p.3)
Q78	“However, our experiments, as shown in Table 7, reveal that FlauBERT is the model with the least word segmentation (1.43 in average), while DrBERT-CP tends to have the highest average segmentation (1.90 in average).” (p.8)
Q79	“Surprisingly, when comparing the performance of these two models on the benchmark tasks, we observe that DrBERT-CP outperforms FlauBERT on 16 out of the 20 tasks, thus contradicting previous conclusions drawn by the community.” (p.8)
Q80	“Yet, tokenization, as it is currently done in MLMs, seems to play a minor role in the performance of systems.” (p.8)
E3C-D8	Azken hilabeteetan Ikernek kodeina + ibuprofenoa hartu ditu, baina ez du hobekuntza handirik nabaritu. — Basque text; completely outside French regulatory deployment scope
E3C-D9	A chest radiograph showed right upper lobe consolidation with volume loss, right para-tracheal and left hilar adenopathy — English clinical text; irrelevant to French deployment
PXCORPUS-D10	/chet — Single-token ASR artifact with no lexical content; not regulatory text
PXCORPUS-D12	i'll agree come on say avec successes merde je vais roulé faut lui faire un et mettre la rame en mode français — English/French code-switching and informal register unlike regulatory prose
PXCORPUS-D13	teralithe 250 milligrammes / le serveur de dialogue met beaucoup de temps à comprendre votre énoncé veuillez reformuler différemment — System dialogue prompt embedded in training example; not a regulatory text pattern
QUAERO-D19	Aucune étude clinique spécifique sur les interactions médicamenteuses n Toutefois, en raison des faibles concentrations plasmatiques du ziconotide… — Truncated mid-sentence artifact from document splitting
QUAERO-D20	r. — Single-character fragment, sentence-splitting artifact

Strengths

- French-only filtering of multilingual corpora ensures language match [Q31]
- Latin script with standard French diacritics matches deployment writing system; no script-level NLP challenges

- HuggingFace toolkit with normalized loaders and predefined splits supports reproducible evaluation [Q48]
- Sentence-splitting addresses long-document input length constraints [Q37], a real challenge for SmPC and CTD documents

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Q37	“Due to the presence of nested entities in annotations, we simplified the evaluation process by retaining only annotations at the higher granularity level from the BigBio (Fries et al., 2022) implementation, following the approach described in Touchent et al. (2023), which translates into an average loss of 6.06% of the annotations on EMEA and 8.90% on MEDLINE. Additionally, considering that some documents from EMEA exceed the maximum input sequence length that most current language models can handle, we decided to split these documents into sentences.” (p.3)
Q48	“We developed a practical toolkit based on the HuggingFace Datasets library (Lhoest et al., 2021). This toolkit includes data loaders that adhere to normalized schemes and predefined data splits. It also provides pre-training and evaluation scripts for each of the tasks, utilizing the HuggingFace Transformers (Wolf et al., 2020) and PyTorch (Paszke et al., 2019) libraries.” (p.5)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions broadly match: French Latin-script text in both contexts. Specific deviation: PxCorpus is transcribed speech with ASR noise (PXCORPUS-D9 'tropatepine chloridrate 1o milligrammes', PXCORPUS-D11 truncations) — distribution mismatch with deployment's structured PDF/Word document input.

IF-2: Yes — deployment is enterprise document management with standard text processing infrastructure; HuggingFace toolkit-based delivery [Q48] is compatible with on-premises or cloud pipelines.

IF-3: Domain-specific form differences: (a) EMEA documents sentence-split [Q37], potentially removing document-level structural context (section 4.1–4.9 hierarchy) that regulatory reasoning depends on; (b) DEFT2020 contains tabular data fragments mixed with prose (CLISTER-D19, CLISTER-D20) that would not occur in deployment's main prose extraction; (c) PxCorpus speech transcripts include code-switching and disfluencies (PXCORPUS-D11, PXCORPUS-D12) entirely absent from regulatory documents.

IF-4: Form mismatches harming external validity: sentence-splitting may hinder evaluation of long-document NLP needed for SmPC/CTD; spoken-language register in PxCorpus does not transfer to regulatory text. Both are partial concerns rather than fatal flaws since the bulk of NER/STS data is written-text French.

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PXCORPUS-D9	tropatepine chloridrate 1o milligrammes 1 comprimé le midi — INN + salt form with minor transcription error ("1o" for "10"), showing ASR noise

PXCORPUS-D 10	/chet — Single-token ASR artifact with no lexical content; not regulatory text
PXCORPUS-D 11	ne pas tenir compte à midi tous les jours merd/ — Profanity and truncation in spoken transcript; absent from regulatory documents
PXCORPUS-D 12	i'll agree come on say avec successes merde je vais roulé faut lui faire un et mettre la rame en mode français — English/French code-switching and informal register unlike regulatory prose
QUAERO-D19	Aucune étude clinique spécifique sur les interactions médicamenteuses n Toutefois, en raison des faibles concentrations plasmatiques du ziconotide... — Truncated mid-sentence artifact from document splitting
CLISTER-D19	Jour 45 - - - - - - - - - 25 µg 0,4 mg" / "Jour 53 15,13 12,0 1,58 - - - - - - 25 µg 0,4 mg — Table row fragments, not natural-language sentences
CLISTER-D20	Jour 91 - - - - - - - - - Fin 0,4 mg" / "Jour 187 1,67 21,2 1,38 - - - - - - - 0,4 mg — Numeric table rows without natural-language predication
QUAERO-D20	r. — Single-character fragment, sentence-splitting artifact

Information Gaps

- No quantification of how often sentence-splitting fragments cross regulatory phrase boundaries (e.g., dose-threshold qualifiers split from drug name).

Remediation

Gap 1: Sentence-splitting of long EMEA documents [Q37] removes document-level structural context (QUAERO-D19, QUAERO-D20) that SmPC/CTD reasoning depends on.

Recommendation: Augment the QUAERO EMEA loader with a section-aware splitter that preserves SmPC section identifiers (4.1–4.9) as input metadata, and evaluate models with and without this structural context to quantify document-level reasoning effects.

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